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Commentary

Unpacking placebo and working memory training effects on cognitive performance



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Footnotes

Author contributions: T.D.W. and H.J. wrote the paper. [AQ1AQ2AQ3AQ4](#)

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See companion article, “Expectation effects in working memory training,” 10.1073/pnas.2209308119.

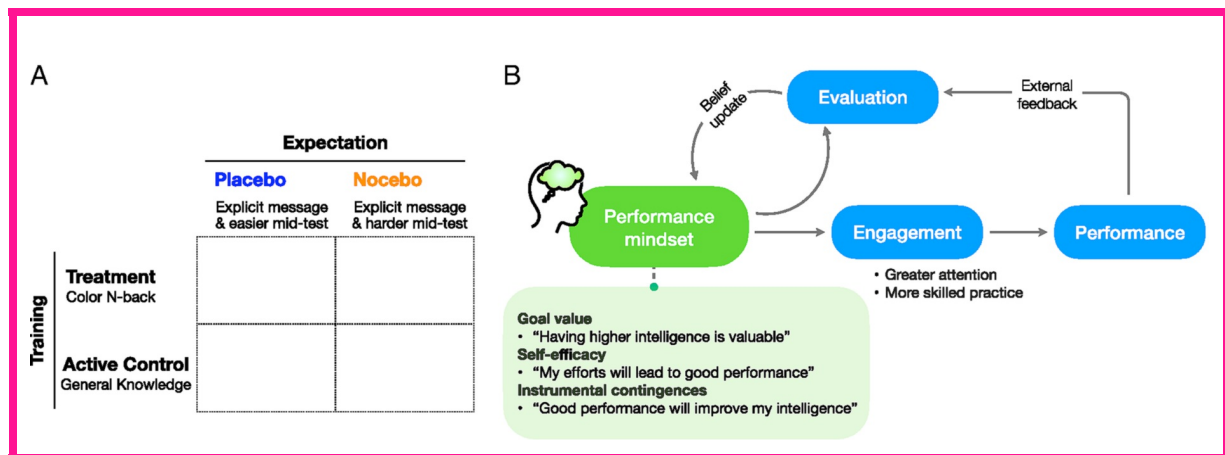
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A fundamental principle of human performance is that it improves with practice (1). There is no doubt that “practice makes perfect” when learning to play the piano, but it also benefits cognitive skills as well, such as improving memory and cognitive control. What is less well understood is whether practice on some tasks, perhaps “brain training” games you can play online, produces gains that transfer to other tasks. It is also unclear what the critical ingredients of practice are. Is it mainly about the amount and quality of practice, or do mindset and motivation also play key roles? Following influential early studies (2), a growing literature suggests that training on working memory (WM) tasks, which challenge the capacity to simultaneously store and manipulate information, can produce sustained benefits on other cognitive control tasks and even perhaps on general fluid intelligence (Gf). Taken at face value, these studies suggest that cognitive training can make one smarter in general and that the key ingredient is practice with specific mental operations. This inference has given rise to a popular wave of commercial brain training products. However, at least two aspects of these studies remain controversial: 1) whether gains transfer to Gf and other cognitive tasks and 2) how much of the effects are attributable to expectations and motivation rather than training per se. In PNAS, Parong et al. (3) address both questions of **training** transfer of **learning** and expectations in a carefully designed randomized, placebo-controlled study.

While it is appealing that training WM could benefit Gf and other cognitive control tasks, extensive literature on skill learning (4) and studies of WM training in particular suggest that transfer is very limited (5, 6). Several meta-analyses do not support transfer to Gf (7); although One meta-analysis reported significant Gf transfer only for studies with passive control conditions (8). However, passive controls leave participants aware of whether they are in a study's active treatment arm or not, and may thus misattribute effects of expectations to training. In addition, Training specificity, rather than transfer, makes sense given that correlations in performance across cognitive control tasks are low (although there are common factors) (9).

~~At the same time,~~ These mixed findings have led researchers ~~have~~ to question whether transfer gains are in fact placebo effects (~~i.e.,~~ effects attributable to expectations and motivation rather than cognitive practice per se). Expectations of reward, punishment, and self-efficacy—one's capacity to act in ways that bring about success—are powerful motivators. Placebo treatments—which induce positive expectations—can increase opioid (10) and dopamine release (11), reduce depression (12), and reduce activation in brain regions related to pain and negative affect. Critically, expectations can also shape what is learned from experience (13) by enhancing reward learning and related activity in dopamine-rich brain regions (14). Inducing positive performance expectations can improve cognitive performance (15) and lead to enhanced gains over the course of WM training (16), although not all studies have found objective performance benefits (17).

To jointly address questions about training and placebo effects, Parong et al. (3) employed a “balanced placebo design,” which crosses a manipulation of an active treatment vs. active control with manipulation of expectations (Fig. 1A) using a 2 training (color *N*-back training as “treatment” vs. trivia training as “active control”) × 2 expectation (placebo vs. nocebo) design. This allowed them to test for effects of training, placebo, and their interaction. Additive effects of training and placebo would suggest that each is independent of the other, whereas interactions would suggest interdependence.



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Note: Figure Replacement

Requested. Fig. 1. Investigating relationships between expectation and performance. (A) The experimental design in Parong et al. (3,12), which crosses a WM training vs. trivia-learning control manipulation with a placebo vs. nocebo expectation manipulation. (B) Potential mechanisms of placebo and nocebo effects. Expectancy manipulations can affect performance mindset—a collection of beliefs about the value and efficacy of training. Mindset can, in turn, affect performance in several ways. First, mindsets can affect attention, task engagement, and task relevance, leading to performance changes. In addition, evaluation of performance feedback (e.g., whether performance meets one's expectations) updates one's mindset. Mindset can affect how performance feedback is evaluated, creating a positive feedback loop that could result in cumulative long-term effects.   

Several features of the study of Parong et al. (undefined) address critiques of some earlier training studies. They randomized assignment to treatment groups, included an active training control group, evaluated posttest gains controlling for baseline performance, and included a reasonably extensive WM training regime. that included Participants underwent 20 training sessions on a color N-back tasks, adaptively adjusting the difficulty level of which was adaptively adjusting to provide a sustained challenge to capacity limits throughout training. A test battery of 10 tasks included a test of “near transfer” to a different letter-based N-back task and “far transfer” to other assessments of cognitive control, attention, and Gf. Finally, they Parong et al. (3) adopted a clever “double-unaware” design using different groups of researchers to deliver the expectancy manipulation and conduct the testing sessions. This way, neither participants nor researchers were aware of the expectancy group assignment during testing, reducing interpersonal transmission of placebo effects (18).

Whereas an earlier study of placebo effects in WM training relied on self-selection into placebo and control groups (16), Parong et al. (3) used a two-part placebo vs. nocebo (negative expectation) manipulation common in studies of placebo analgesia. First, they provided pretraining instructions that emphasized transfer in the placebo group and lack of transfer in the nocebo group. Second, they provided a midtraining manipulation designed to associatively reinforce initial beliefs. In this session, the positive and negative expectancy groups performed easier and more difficult tasks, respectively. As in previous studies of placebo effects on pain, this gave

participants a “lived experience” of benefits that match their initial expectations, which may be critical for biasing learning toward initial expectations (13). Perhaps crucially, the difficulty reduction in the positive expectancy group was calibrated to produce an effect on performance that was noticeable but not obviously detectable as a change in the task itself, leading participants to attribute their good (or poor) performance to the training.

The results were surprising in several respects. On the primary posttraining test, performance across the 10 test tasks as a whole was higher in the WM training groups than control training and higher in the placebo than nocebo groups, with no significant interaction. This supports the idea that training and expectation both contribute to performance independently. However, post hoc tests on individual tasks revealed an interesting pattern. As expected, near-transfer performance to the letter *N* back improved with WM training—and also, interestingly, with placebo (vs. compared with nocebo). Effects across other individual tasks were sparse, with significant WM training improvements in one of the nine far-transfer tasks and placebo effects in three tasks. This pattern suggests that the overall gains across tasks might reflect weak, broadly distributed effects of both manipulations. Although the sample size was relatively large considering the longitudinal nature of the study (125 participants completed the 20 trainings and all assessments, with about $n = 30$ per group), power to detect individual task effects is likely to be low, and larger studies are needed to ascertain the pattern of which types of tasks benefit more from WM training and which do not.

This caveat notwithstanding, a particularly suggestive pattern emerged when considering effects on Gf tasks. These tasks, including the Raven’s Progressive Matrices task and UC-Matrix Reasoning University of California Matrix Reasoning Task AQ5 Test, are correlated with performance on a wide variety of tasks and are standard measures of Gf. These Gf measures only showed an effect of placebo vs. nocebo suggestions and no significant training effect [table 1 in Parong et al. (3)]. On one hand, this is consistent with studies finding that far-transfer effects are limited. On the other hand, it provides a provocative suggestion that widely used measures of cognitive ability can be influenced by participants’ beliefs. ~~about their performance and perhaps, their own self-efficacy.~~

How might placebo and nocebo suggestions affect objective cognitive performance? The decision to engage in a task (and how vigorously we do so) is a complex one that is determined by multiple beliefs: belief that the end goal is valuable (“goal valence”), that engagement and effort will lead to good performance (“self-efficacy”), and that good performance will lead to valued outcomes [“instrumental contingencies” (19)]. Together, these beliefs might be called a “performance mindset” (Fig. 1B). Placebo and nocebo suggestions can affect each of these motivational factors. In addition, placebo effects on one’s performance mindset might affect how we evaluate objective feedback, which in turn, informs how the mindset is updated. Such recursive positive feedback loops can compound the effects of initial suggestions over time, creating a virtuous cycle of encouragement and grit AQ6 or a vicious cycle of negative evaluation of the task and oneself.

The study of Parong et al. (3) was not designed to unpack these processes, and there is much to explore at both psychological and neural levels. One suggestive finding is that the individuals most susceptible to placebo and nocebo effects were those highest in growth mindset—the belief that intelligence is malleable and can improve with training. Growth mindset is closely related to self-efficacy, a concept similar to ~~what motivation theorist Victor Vroom (19) called “expectancy”~~. ~~and it is possible that~~ Those highest in it ~~were~~ may be most open to the suggestions that training ~~would~~ will have potent effects for better or worse. However, other findings suggest complex

patterns and the need for new measures and larger studies. For example, positive feedback between expectations and self-evaluated performance implies synergy between the placebo manipulation and training, resulting in the largest training gains for the placebo group (an “overadditive” interaction). Parong et al. (3) do not find such an interaction. In fact, placebo effects are numerically about twice as large in the control training than WM training condition (an “underadditive” interaction). This suggests that either placebo or training may be sufficient to produce gains, although larger studies are needed for sufficient power to test this effect. In addition, while some studies of placebo analgesia have found that the strongest placebo responders are high in “behavioral activation” (e.g., reward seeking and fun seeking) and optimism (20), in the study by Parong et al. (3), the strongest placebo effects were found in those who were lowest in behavioral activation and extraversion. This pattern suggests that placebo and nocebo suggestions may compensate for low intrinsic engagement. More broadly, which trait measures predict placebo and nocebo effects may be complex and situation dependent.

In sum, while the pattern of their weave may be complex, expectations and learning are fundamentally entangled. Performance on any task requires motivation, which is grounded in beliefs as well as experiences of success. Practice serves not only to build skills but also, to provide critical input into our ongoing assessment of what is possible and what is valuable. Researchers, educators, and cognitive trainees alike would do well to consider both aspects in designing training programs. The study by Parong et al. (3) provides a rare and welcome inroad into this vast frontier.

References

1. M. Ebbinghaus, *A Contribution to Experimental Psychology, 1913* (Teachers College, Columbia University, New York, NY, 1885). ↑
2. S. M. Jaeggi, M. Buschkuhl, J. Jonides, W. J. Perrig, Improving fluid intelligence with training on working memory. *Proc. Natl. Acad. Sci. U.S.A.* **105**, 6829–6833 (2008). ↑
3. J. Parong, A. R. Seitz, S. M. Jaeggi, C. S. Green, Expectation effects in working memory training. *Proc. Natl. Acad. Sci. U.S.A.* **119**, e2209308119 (2022). ↑
4. A. F. Healy, E. L. Wohldmann, “Specificity and transfer of learning” in *The Psychology of Learning and Motivation*, B. H. Ross, Ed. (Elsevier Academic Press, 2012), pp. 227–253. ↑
5. A. M. Owen et al., Putting brain training to the test. *Nature* **465**, 775–778 (2010). ↑
6. T. L. Harrison et al., Working memory training may increase working memory capacity but not fluid intelligence. *Psychol. Sci.* **24**, 2409–2419 (2013). ↑
7. A. Soveri, J. Antfolk, L. Karlsson, B. Salo, M. Laine, Working memory training revisited: A multi-level meta-analysis of n-back training studies. *Psychon. Bull. Rev.* **24**, 1077–1096 (2017). ↑
8. J. Au et al., Improving fluid intelligence with training on working memory: A meta-analysis. *Psychon. Bull. Rev.* **22**, 366–377 (2015). ↑

9. A. Miyake *et al.*, The unity and diversity of executive functions and their contributions to complex “Frontal Lobe” tasks: A latent variable analysis. *Cognit. Psychol.* 41, 49–100 (2000). 
10. T. D. Wager, D. J. Scott, J.-K. Zubieta, Placebo effects on human μ -opioid activity during pain. *Proc. Natl. Acad. Sci. U.S.A.* 104, 11056–11061 (2007). 
11. R. de la Fuente-Fernández *et al.*, Expectation and dopamine release: Mechanism of the placebo effect in Parkinson’s disease. *Science* 293, 1164–1166 (2001). 
- ~~12. B. R. Rutherford, S. P. Roose, A model of placebo response in antidepressant clinical trials. *Am. J. Psychiatry* 170, 723–733 (2013).   ~~
12. B. R. Rutherford *et al.*, Patient Expectancy as a Mediator of Placebo Effects in Antidepressant Clinical Trials. *Am. J. Psychiatry*. 174, 135–142 (2017).   
13. M. Jepma, L. Koban, J. van Doorn, M. Jones, T. D. Wager, Behavioural and neural evidence for self-reinforcing expectancy effects on pain. *Nat. Hum. Behav.* 2, 838–855 (2018). 
14. L. Schmidt, E. K. Braun, T. D. Wager, D. Shohamy, Mind matters: Placebo enhances reward learning in Parkinson’s disease. *Nat. Neurosci.* 17, 1793–1797 (2014). 
15. B. Shiv, Z. Carmon, D. Ariely, Placebo effects of marketing actions: Consumers may get what they pay for. *J. Mark. Res.* 42, 383–393 (2005). 
16. C. K. Foroughi, S. S. Monfort, M. Paczynski, P. E. McKnight, P. M. Greenwood, Placebo effects in cognitive training. *Proc. Natl. Acad. Sci. U.S.A.* 113, 7470–7474 (2016). 
17. A. Looby, M. Earleywine, Expectation to receive methylphenidate enhances subjective arousal but not cognitive performance. *Exp. Clin. Psychopharmacol.* 19, 433–444 (2011). 
18. P. A. Chen *et al.*, Socially transmitted placebo effects. *Nat. Hum. Behav.* 3, 1295–1305 (2019). 
19. V. H. Vroom, *Work and Motivation* (John Wiley and Sons, Inc., 1964). 
20. P. Schweinhardt, D. A. Seminowicz, E. Jaeger, G. H. Duncan, M. C. Bushnell, The anatomy of the mesolimbic reward system: A link between personality and the placebo analgesic response. *J. Neurosci.* 29, 4882–4887 (2009). 

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